OPTICAL MEMORY BUS AND INTERFACE FOR HIGH-BANDWIDTH MEMORY ACCESS

Reinaldo Javier Aguiar Marcano

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Inventor: Reinaldo Javier Aguiar Marcano, a citizen of the United States, residing at 2302 Britton Ridge Drive, Katy, TX 77494.

Abstract

A system and method for an ultra-high-bandwidth memory interface that replaces conventional electrical pin-based connectors with a massively parallel optical interconnect. The system comprises a memory module (e.g., a DIMM) equipped with a high-density array of photon sensors and a corresponding motherboard memory slot equipped with a high-density array of photon emitters. Data is transferred between the processor and the memory module as a sequence of two-dimensional frames of light, where each pixel in the frame acts as an individual data channel. This architecture provides a memory bus with a theoretical bandwidth in the petabit-per-second (Pbps) range, effectively eliminating the "memory wall" bottleneck that limits the performance of modern processors in data-intensive applications.

Background of the Invention

For decades, the performance of computer systems has been limited by the "memory wall"—the growing disparity between the speed of processors and the speed at which they can access data from main memory (RAM). While CPU clock speeds and core counts have increased exponentially, the bandwidth of the electrical memory bus that connects the CPU to the RAM modules has not kept pace. This memory bottleneck is a primary performance-limiting factor in fields such as artificial intelligence, high-performance computing (HPC), and large-scale data analysis, where processors are often starved for data. Conventional memory interfaces, such as the Dual In-Line Memory Module (DIMM), rely on hundreds of metallic pins, an architecture that is reaching its physical limits due to issues of signal integrity, electromagnetic interference, and power consumption at higher speeds. A new paradigm is needed to provide processors with the memory bandwidth they require.

Summary of the Invention

The present invention demolishes the memory wall by replacing the physical-contact electrical memory bus with a massively parallel, free-space optical interconnect. The invention, referred to as the Optical Memory Bus, applies the principles of frame-based optical data transmission to the CPU-to-memory interface.

The system comprises a memory module, analogous to a DIMM, that has its traditional edge connector of metallic pins replaced with a compact, high-density array of photon sensors (e.g., a CMOS sensor). The corresponding memory slot on the computer's motherboard is equipped with a matching array of individually addressable photon emitters (e.g., a micro-OLED or micro-LED array).

When the memory module is installed, the sensor array on the module and the emitter array in the slot are precisely aligned. Data is transferred not as a serial or limited-parallel electrical signal, but as a sequence of two-dimensional frames of light. The system's processor encodes data into these frames, where each pixel's color and brightness represents a packet of data. With millions of pixels transmitting in parallel at a high frequency (e.g., 1 MHz), the resulting aggregate bandwidth can reach the petabit-per-second scale, effectively giving the processor near-instantaneous access to main memory.

The invention includes methods for ensuring robust connection despite the microscopic

tolerances, such as using "channel elements" (grouping multiple physical pixels for one data channel) to increase alignment tolerance, and an electronic handshake protocol for self-calibration upon module installation.

Statement of License Grant to the United States Government

The inventor, Reinaldo Javier Aguiar Marcano, hereby grants to the United States Government and its agencies a perpetual, royalty-free, non-exclusive license to practice and use the invention described herein for any and all governmental and non-commercial purposes.

Brief Description of the Drawings

FIG. 1 is a conceptual diagram of a computer motherboard showing a CPU with an optical socket and its connection to a memory module via the Optical Memory Bus.

FIG. 2 is a detailed, exploded view of the optical interface, showing the emitter array on the motherboard slot and the sensor array on the memory module.

FIG. 3 is a block diagram illustrating the data flow from the CPU to the memory module across the optical bus.

Detailed Description of the Invention

Referring to **FIG. 1**, a computer system motherboard (101) is shown with a processor (CPU) (102) installed. The CPU (102) communicates with the rest of the system via an optical interface (e.g., a "CPU Light Socket"). The motherboard (101) includes one or more memory slots (103) which constitute the host-side of the Optical Memory Bus. A memory module (104), analogous to a standard DIMM, is installed into one of these slots. The connection between the memory slot (103) and the memory module (104) is entirely optical. The data pathway (105) between the CPU and memory is therefore a high-bandwidth optical link, free from the constraints of an electrical bus.

FIG. 2 provides an exploded view of the optical interface. The motherboard memory slot (103) contains a high-density array of photon emitters (201), such as a micro-LED or micro-OLED panel. It also contains an interlocking mechanism (202) for precise alignment. The

memory module (104) has its traditional electrical pins replaced with a high-density array of photon sensors (203), such as a CMOS image sensor, and a corresponding interlocking mechanism (204). When the module is installed, the two arrays (201, 203) are brought into close, parallel alignment within a shielded enclosure that blocks ambient light.

The data transmission process is illustrated in **FIG. 3**. A memory controller (301), which may be part of the CPU or a separate chipset, sends a data request. This request is processed by an encoder (302), which translates the data to be written to or read from memory into a sequence of light frames. The encoder drives the emitter array (201) to produce these frames. For a 'write' operation, data from the CPU is encoded into the light frames. For a 'read' operation, the emitter array might send a frame of 'address' information to the memory module. The light frames are captured by the sensor array (203) on the memory module. The data is decoded by a decoder circuit (303) on the module and passed to the actual memory chips (DRAM) (304). In a read operation, the process is reversed, with the memory module using its own emitter array (not shown) and the motherboard slot its own sensor array (not shown).

The immense bandwidth is achieved through massive parallelism. An interface area of 25mm x 6mm with a 2-micrometer pixel pitch yields approximately 37.5 million parallel channels. With each channel encoding 24 bits of data per cycle and operating at a conservative 1 MHz, a single memory module can achieve a theoretical bandwidth of 0.9 petabits per second (Pbps).

Claims

What is claimed is:

1. A memory interface system for a computer, comprising:

    - a memory slot on a motherboard, said slot containing a first two-dimensional array of individually addressable electromagnetic radiation emitters; and
    - a removable memory module, said module containing memory chips and a second two-dimensional array of electromagnetic radiation sensors, wherein said second array is positioned to optically couple with the first array when the module is installed in the slot.

2. The system of claim 1, wherein data is transmitted between the motherboard and the

memory module as a sequence of frames, with each frame comprising a plurality of data packets encoded as the emission characteristics of the emitters in the first array.

3. The system of claim 2, wherein the emission characteristics include at least one of wavelength or intensity.

4. The system of claim 1, wherein the emitters are micro-LEDs or micro-OLEDs and the sensors are a CMOS image sensor.

5. The system of claim 1, wherein the aggregate data transmission bandwidth of the interface is in the terabit-per-second or petabit-per-second range.

6. A memory module for a computer system, comprising:

   - a plurality of memory integrated circuits (chips);
   - a printed circuit board on which the memory chips are mounted; and
   - a two-dimensional array of photon sensors integrated onto the module, configured to serve as the primary data interface for the module.

7. The memory module of claim 6, wherein the array of photon sensors replaces a traditional metallic edge connector.

8. A method for accessing computer memory, comprising the steps of:

   - generating, by a memory controller, a sequence of data frames, wherein data to be written to a memory module is encoded into the emission characteristics of a two-dimensional array of emitters located in a motherboard memory slot;
   - transmitting said sequence of frames as light from the emitter array;
   - capturing, by a sensor array on the memory module, said sequence of frames; and
   - decoding said frames and writing the data to memory chips on the module.

9. The method of claim 8, wherein the data is transferred at a rate exceeding 1 terabit per second.

10. The method of claim 8, further comprising a self-calibration step upon installation of the module, wherein the memory module and the memory slot electronically negotiate a data map to optimize accuracy.

Drawings

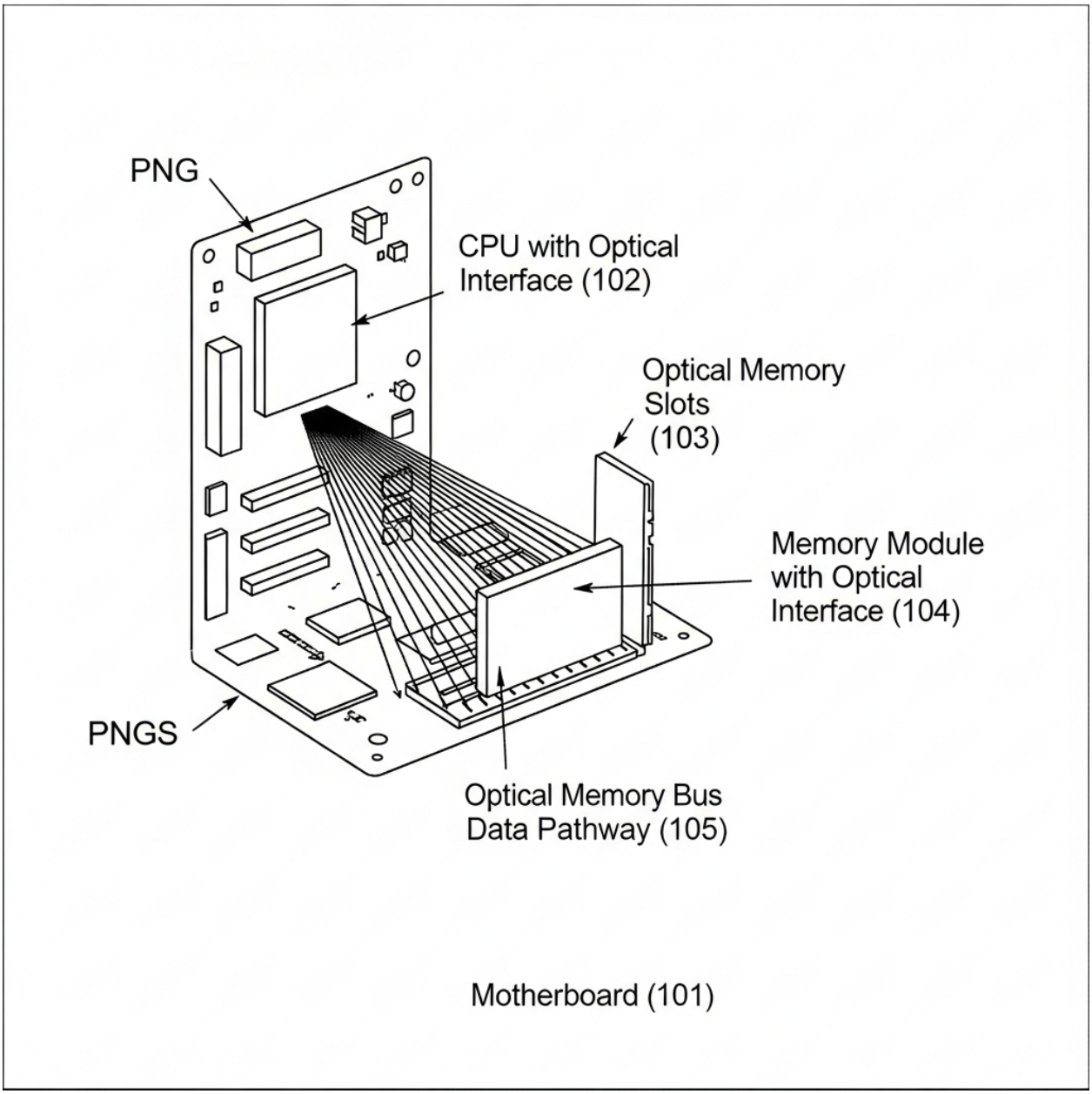


Figure 1: A conceptual diagram of a computer motherboard showing a CPU with an optical socket and its connection to a memory module via the Optical Memory Bus.

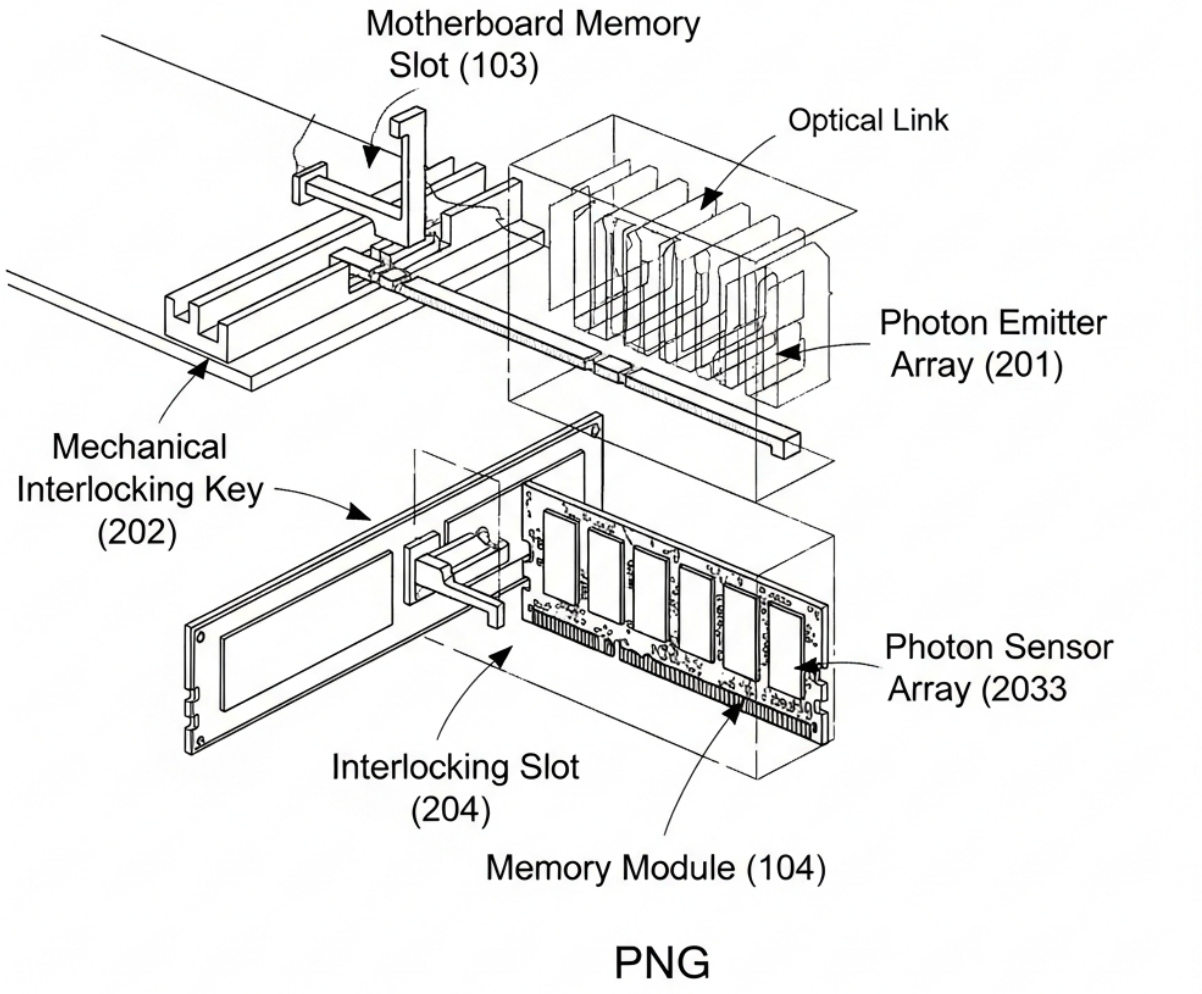


Figure 2: A detailed, exploded view of the optical interface, showing the emitter array on the motherboard slot and the sensor array on the memory module.

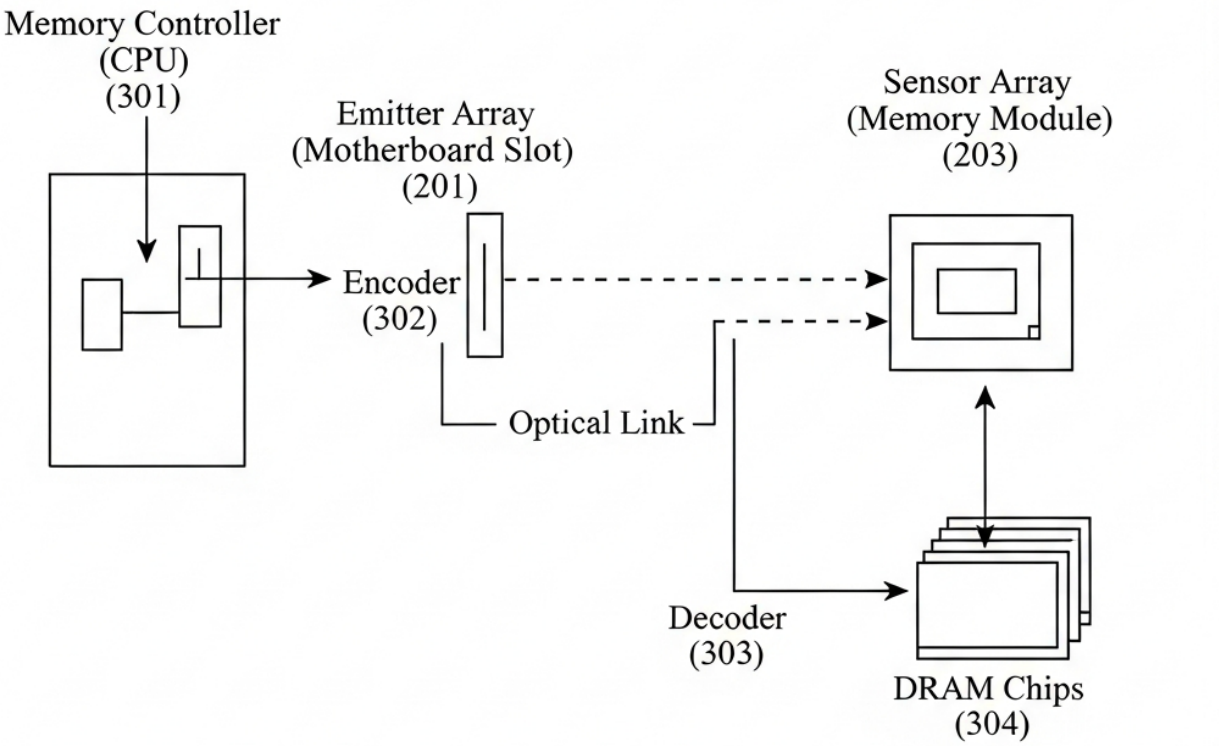


Figure 3: A block diagram illustrating the data flow from the CPU to the memory module across the optical bus.